

Deriv

$$v = \frac{dx}{dt}$$

$$\frac{d(f+g)}{dx} = \frac{df}{dx} + \frac{dg}{dx}$$

$$\frac{d(cf)}{dx} = c \frac{df}{dx}$$

$$\frac{d(g \cdot f)}{dx} = \frac{df}{dx} g + f \frac{dg}{dx}$$

$$\frac{d\left(\frac{f}{g}\right)}{dx} = \frac{\frac{df}{dx} g - f \frac{dg}{dx}}{g^2}$$

$$\frac{d(f(g(x)))}{dx} = \frac{df(g)}{dx} \cdot \frac{dg}{dx}$$

Integral

$$x = \int_1^2 v(t) dt$$

| $f(x)$        | $\frac{df(x)}{dx}$ | $\int$                    |
|---------------|--------------------|---------------------------|
| $c$           | $0$                | $cx + c_1$                |
| $x^n$         | $nx^{n-1}$         | $\frac{x^{n+1}}{n+1} + c$ |
| $\frac{1}{x}$ | $-x^{-2}$          | $\ln x  + c$              |
| $\sin x$      | $\cos x$           | $-\cos x + c$             |
| $\cos$        | $-\sin x$          | $\sin x + c$              |

|         |               |                   |
|---------|---------------|-------------------|
| $e^x$   | $e^x$         | $e^x + c$         |
| $\ln x$ | $\frac{1}{x}$ | $x \ln x - x + c$ |

1.1

$$t = 4 \text{ s}$$

$$c_2 = 340 \frac{\text{m}}{\text{s}}$$

$$s = ?$$



$$s = c_2 t = \underline{1360 \text{ m}}$$

(zanemarimo hitrost svetlobe)

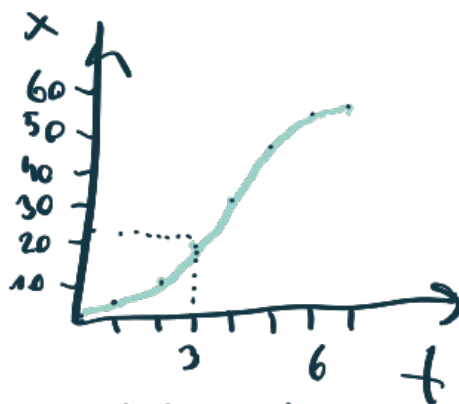
1.2

|               |   |   |    |    |    |    |    |    |
|---------------|---|---|----|----|----|----|----|----|
| $t[\text{s}]$ | 0 | 1 | 2  | 3  | 4  | 5  | 6  | 7  |
| $x[\text{m}]$ | 0 | 4 | 11 | 21 | 35 | 45 | 53 | 57 |

$$\bar{v}_{1,2} = ?$$

$$t_1 = 3 \text{ s}$$

$$t_2 = 6 \text{ s}$$



$$\bar{v}_6 = \frac{x(7) - x(5)}{2 \Delta t} = \underline{6 \frac{\text{m}}{\text{s}}}$$

$$a_6 = -4 \frac{\text{m}}{\text{s}^2}$$

$$\bar{v}_{23} = \frac{21 \text{ m} - 11 \text{ m}}{1 \text{ s}} = 10 \frac{\text{m}}{\text{s}}$$

$$\bar{v}_{34} = 14 \frac{\text{m}}{\text{s}}$$

$$\bar{v}_3 = 12 \frac{\text{m}}{\text{s}}$$

$$a_3 = \frac{v_{34} - v_{23}}{\Delta t} = \underline{4 \frac{\text{m}}{\text{s}^2}}$$

1.3

|     |   |   |   |   |
|-----|---|---|---|---|
| $t$ | 0 | 1 | 2 | 3 |
|-----|---|---|---|---|

| $v$ | 1 | 2     | 3     |       |
|-----|---|-------|-------|-------|
| $x$ | 0 | $x_1$ | $x_2$ | $x_3$ |

določí  $x_1, x_2, x_3$

$$x_1 = \Delta t v_1 = 1$$

$$x_2 = x_1 + \Delta t v_2 = 3$$

$$x_3 = x_2 + \Delta t v_3 = 6$$

1.4

$$x(t) = At^2 - Bt^3$$

$$A = 3 \frac{\text{m}}{\text{s}^2}$$

$$B = 2 \frac{\text{m}}{\text{s}^3}$$

$$t = [0,6\text{s}; 0,8\text{s}]$$

$$\bar{v}, \bar{a} = ?$$

$$v_{0,7\text{s}}, a_{0,7\text{s}} = ?$$

$$v(t) = \frac{dx}{dt} = 2At - 3Bt^2$$

$$a(t) = \frac{dv}{dt} = 2A - 6Bt$$

$$\textcircled{1} v_{0,7}, a_{0,7}$$

$$v_{0,7} = 6 \cdot 0,7 \frac{\text{m}}{\text{s}^2} \cdot 0,7 - 6 \cdot 0,49 \frac{\text{m}}{\text{s}^3} \cdot 0,7^2 = 1,26 \frac{\text{m}}{\text{s}}$$

$$a_{0,7} = 6 \frac{\text{m}}{\text{s}^2} - 12 \cdot 0,7 \frac{\text{m}}{\text{s}^3} = -2,4 \frac{\text{m}}{\text{s}^2}$$

$$\textcircled{2} \bar{v} = \frac{\Delta x}{\Delta t} = \frac{\int_{t_1}^{t_2} v dt}{t_2 - t_1} =$$

$$= \frac{x_2 - x_1}{t_2 - t_1} = 1,24 \frac{\text{m}}{\text{s}}$$

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} = -2,4 \frac{\text{m}}{\text{s}^2}$$

a linearno  
naraščajoča, zato  
je enak,  
v pa naraščajoča

s kvadratom,  
zato  $\ddot{v} \neq v, r$

1.5

$$v(t) = 2At - 3Bt^2$$

$$A = 3 \frac{\text{m}}{\text{s}^2}$$

$$B = 2 \frac{\text{m}}{\text{s}^3}$$

$$x(t_1 = 0,9\text{s}) = ?$$

$$x_0 = 0$$

$$x = \int_0^{t_1} v(t) dt = 2A \frac{t^2}{2} - 3B \frac{t^3}{3} = 3 \frac{\text{m}}{\text{s}^2} \cdot 0,9^2 \text{s}^2 - 2 \frac{\text{m}}{\text{s}^3} (0,9\text{s})^3 =$$

$$\underline{\underline{= 1\text{m}}}$$

1.6 Z 9 nal. 6

$$v = k\sqrt{t}$$

$$k = 5 \frac{\text{m}}{\text{s}^{\frac{3}{2}}}$$

$$t_1 = 30\text{s} \quad t_2 = 0$$

$$x_1, a_1 = ?$$

$$a_2 = ?$$

$$x_1 = \int_0^{t_1} v dt = k \int_0^{t_1} \sqrt{t} dt = \left( \frac{2}{3} k t^{\frac{3}{2}} \right) \Big|_0^{t_1} =$$

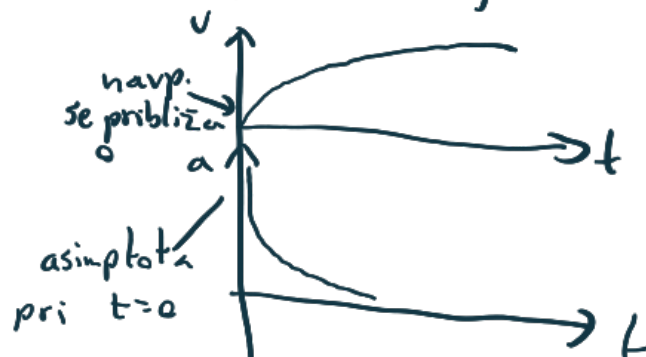
$$= \frac{2}{3} \cdot 5 \frac{\text{m}}{\text{s}^{\frac{3}{2}}} \cdot 30^{\frac{3}{2}} \cancel{\text{s}^{\frac{3}{2}}} = \underline{\underline{548\text{m}}}$$

$$a = \frac{dv}{dt} = k \frac{d(t^{\frac{1}{2}})}{dt} = \frac{1}{2} k t^{-\frac{1}{2}}$$

$$a_1 = \frac{1}{2} \cdot 5 \frac{\text{m}}{\text{s}^{\frac{3}{2}}} \cdot \frac{1}{\sqrt{30} \text{s}^{\frac{1}{2}}} = \underline{\underline{0,46 \frac{\text{m}}{\text{s}^2}}}$$

$$a_2 = \frac{5 \text{ m}}{2 \text{ s}^{\frac{3}{2}} \cdot \sqrt{0}} \rightarrow \infty$$

Posp. ob  $t=0$  je neskončen:



1.7 29 nal. 5

$$v_0 = 4 \frac{\text{m}}{\text{s}}$$

$$t_1 = 10 \text{ s}$$

$$a = -k v^2$$

$$k = 0,65 \text{ m}^{-1}$$

$$v_1, x_1 = ?$$

ali se ustavi?

Ne, ker  $\ln v$   
enačbi za pot  
ves čas  
navaščena.

$$a = \frac{dv}{dt} = -k v^2$$

$$-k \int_0^t dt = \int_{v_0}^v \frac{dv}{v^2}$$

$$-k(t-0) = \left( -1 v^{-1} \right) \Big|_{v_0}^v = \frac{1}{v_0} - \frac{1}{v}$$

$$kt = \frac{1}{v} - \frac{1}{v_0}$$

$$\frac{1}{v} = kt + \frac{1}{v_0}$$

$$v = \left( kt + \frac{1}{v_0} \right)^{-1} = \frac{v_0}{ktv_0 + 1}$$

$$v_1 = \left( 0,65 \cdot 10 \frac{\text{s}}{\text{m}} + \frac{1}{4 \frac{\text{s}}{\text{m}}} \right)^{-1} = \left( 6,75 \frac{\text{s}}{\text{m}} \right)^{-1} = \underline{\underline{0,148 \frac{\text{m}}{\text{s}}}}$$

$$x = \int_0^t \left( kt + \frac{1}{v_0} \right)^{-1} dt = \int_0^t \frac{\overbrace{v_0}^{\text{konst.}}}{\underbrace{ktv_0 + 1}_{\text{konst.}}} dt =$$

$$= \frac{1}{k} \int_1^{ktv_0 + 1} \frac{1}{u} \cdot \frac{du}{k v_0} =$$

$$= \frac{1}{k} \left( \ln |u| \right) \Big|_1^{ktv_0 + 1} = \frac{1}{k} \left( \ln |ktv_0 + 1| - \ln |1| \right) =$$

$$= \frac{1}{k} \left( \ln \left| \frac{ktv_0 + 1}{1} \right| \right)$$

$$x_1 = \frac{1 \text{ m}}{0,65} \cdot \ln \left| 0,65 \frac{1}{\text{m}} \cdot 10 \text{ s} \cdot 4 \frac{\text{m}}{\text{s}} + 1 \right| = \underline{\underline{5,1 \text{ m}}}$$

1.8

$$l = 2 \text{ m}$$

$$\Delta t = 0,25 \text{ s}$$



$$g = 10 \frac{\text{m}}{\text{s}^2}$$

$$x = x_0 + v_0 t + \frac{g t^2}{2}$$

$$\int v dt = v + at$$

$h = ?$



$$t_2, v_2$$

$$x = \int_{x_0}^x v dt = x_0 + v_0 t + \frac{g t^2}{2}$$

$$\bar{v} = \frac{l}{\Delta t} = 8 \frac{m}{s}$$

$$v_1 = v_0 - g \cdot \frac{\Delta t}{2} = 6,75 \frac{m}{s}$$

$$v_1 = g t_1; \quad h = \frac{g t_1^2}{2} = \frac{v_1^2}{2g} = \underline{2,3 m}$$

$$t_1 = \frac{v_1}{g}$$

SAMO ZA  $a = konst.$ :

$$a = \frac{dv}{dt} \frac{dx}{dx} = v \frac{dv}{dx}$$

$$\int a dx = \int v dv$$

$$ax = \left( \frac{v^2}{2} + C \right) \Big|_0^v = \frac{v^2 - v_0^2}{2}$$

$$v = \sqrt{2ax + v_0^2}$$

1.9) 29 nal. 12 (spustimo kamen u vodnjak, po  $t_1$  zaslišimo pljusk)

$$\frac{t_1 = 5 s}{c_2 = 340 \frac{m}{s}}$$

$h = ?$



$$h = \frac{g t_1'^2}{2}$$

$$h = c_2 \cdot t'' = c_2 (t_1 - t')$$

$$\frac{g t_1'^2}{2} = c_2 (t_1 - t')$$

$$g t_1'^2 + 2 c_2 t' - 2 c_2 t_1 = 0$$

$$t'_1 = 4,678 \text{ s}$$

$$t'_2 = -72,678 \text{ s} \quad \times$$

$$h = c_2(t_1 - t'_1) = \underline{\underline{109 \text{ m}}}$$

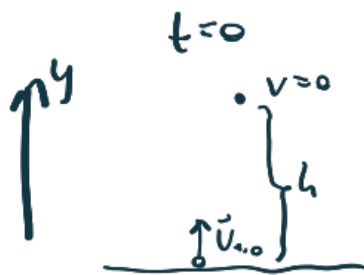
1.10 29 ml. 7

$$V_{10} = 10 \text{ m/s}$$

$$V_{2.0} = 0$$

$$h = ?$$

$$\Delta t = t_3 - t_2 = ?$$



$t_3$

oo

$t_1$

$t_2$



2. kamen:

$$y_2 = h - \frac{gt^2}{2}$$

1. kamen:

$$y_1 = v_0 t - \frac{gt^2}{2}$$

$t_2$ :

$$y_2 = 0 \Rightarrow 0 = h - \frac{gt_2^2}{2}$$

$$t_2 = \sqrt{\frac{2h}{g}} = \sqrt{2} \text{ s}$$

$$t_1: h - \frac{gt^2}{2} = v_0 t - \frac{gt^2}{2}$$

$$h = v_0 t_1$$

$$h - \frac{gt^2}{2} = \frac{h}{2}$$

$$h = \frac{gt_1^2}{2}$$

$$gt_1^2 - v_0 t_1 = 0$$

$$t_1 (gt_1 - v_0) = 0$$

$$t_1 = \frac{v_0}{g} = 1 \text{ s}$$

$$h = v_0 t_1 = \underline{\underline{10 \text{ m}}}$$

$$\Delta t = 2 \text{ s} - \sqrt{2} \text{ s} = \underline{\underline{0,6 \text{ s}}}$$

$t_3$ :

$$y_1 = 0 \Rightarrow v_0 t_3 - \frac{gt_3^2}{2} = 0$$

$$2v_0 t_3 - gt_3^2 = 0$$

$$t_3 (2v_0 - gt_3) = 0$$

$$t_3 = \frac{2v_0}{g} = 2 \text{ s}$$

(1.)

1

1.1.2

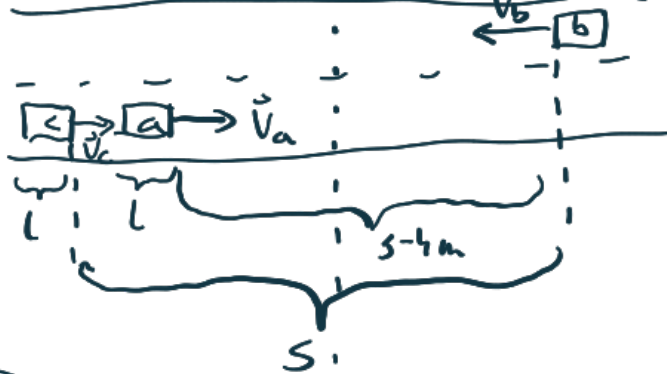
$$V_a = 80 \frac{\text{km}}{\text{h}} = 22,2 \frac{\text{m}}{\text{s}}$$

$$S = 160 \text{ m}$$

$$V_b = V_a = V_c$$

$$l = 4 \text{ m}$$

$$a_i = ?$$



2.



Pred trkom:

$$a: s_a = \frac{(S-l)}{2} = 76 \text{ m}$$

$$b: s_b = \frac{(S-l)}{2} = 76 \text{ m}$$

$$c: s_c = s_a + 2l = 84 \text{ m}$$

$$s_a = V_a t$$

$$t = \frac{s_a}{V_a} = \frac{76 \text{ m} \cdot \text{s}}{22,2 \text{ m}} = 3,42 \text{ s}$$

$$s_c = V_a t + \frac{at^2}{2}$$

$$at^2 = s_c - 2V_a t$$

$$a = \frac{s_c - 2V_a t}{t^2} = 1,4 \frac{\text{m}}{\text{s}^2}$$

2.1

$$h, l, v_0$$

$$\varphi, \ominus$$



$$x_i = v_{xi} \cdot t = v_0 t \cos \varphi$$

$$y_i = v_{yi} \cdot t - \frac{gt^2}{2} = v_0 t \sin \varphi - \frac{gt^2}{2}$$

$$x_t = l$$

$$y_t = h - \frac{gt^2}{2}$$

$$v = v$$

$$v_i = v_i$$

$$x_i = x_t$$

$$L = v_0 t \cos \varphi$$

$$y_i = y_t$$

$$v_0 t \sin \varphi - \frac{gt^2}{2} = h - \frac{gt^2}{2}$$

$$h = v_0 t \sin \varphi$$

$$\frac{h}{L} = \frac{v_0 t \sin \varphi}{v_0 t \cos \varphi} = \tan \varphi$$

$$\varphi = \arctan \frac{h}{L}$$



$$x_i: v_x = v_0 \cos \varphi$$

$$y_i: v_y = v_0 \sin \varphi - gt$$

$$= v_0 \sin \varphi - \frac{gh}{v_0 \sin \varphi}$$

$$t = \frac{h}{v_0 \sin \varphi}$$

$$\tan \Theta = \frac{v_0 \sin \varphi - \frac{gh}{v_0 \sin \varphi}}{v_0 \cos \varphi}$$

$$= \tan \varphi - \frac{gh}{v_0^2 \cdot \frac{1}{2} \sin 2\varphi}$$

$$= \tan \varphi - \frac{2gh}{v_0^2 \sin 2\varphi}$$



$$\Theta = \arctan \left( \tan \varphi - \frac{2gh}{v_0^2 \sin 2\varphi} \right)$$

kat, pod katerim zadane tarzo

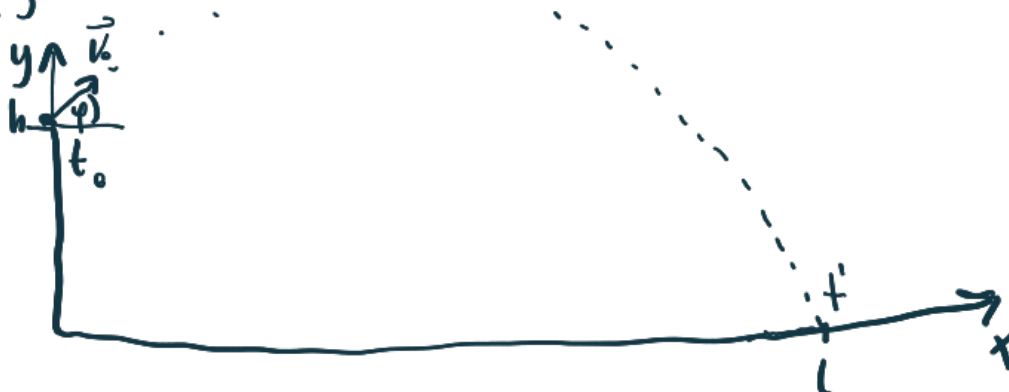
2.2 29 / 25

$$v_0 = 10 \frac{m}{s}$$

$$h = 12 m$$

$$L = \max$$

$$\varphi = ?$$



$$x = v_x t = v_0 t \cos \varphi$$

$$y = h + v_y t - \frac{gt^2}{2} = h + v_0 t \sin \varphi - \frac{gt^2}{2}$$

$$t = t' : x = v_0 t' \cos \varphi = l$$

$$y = h + v_0 t' \sin \varphi - \frac{g t'^2}{2} = 0$$

$$t' = \frac{l}{v_0 \cos \varphi}$$

$$Q = h + \frac{v_0 \sin \varphi}{v_0 \cos \varphi} - \frac{g l^2}{v_0^2 \cos^2 \varphi \cdot 2} = h + l \cdot \tan \varphi - \frac{g l^2}{2 v_0^2 \cos^2 \varphi}$$

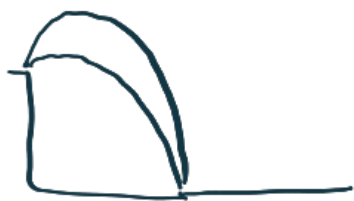
$$\cos^2 \varphi + \sin^2 \varphi = 1 \quad / : \cos^2 \varphi$$

$$1 + \tan^2 \varphi = \frac{1}{\cos^2 \varphi}$$

$$Q = h + l \tan \varphi - \frac{g l^2 (1 + \tan^2 \varphi)}{2 v_0^2} = - \frac{g l^2 \tan^2 \varphi}{2 v_0^2} + (l \tan \varphi + h - \frac{g l^2}{2 v_0^2})$$

$$\tan \varphi_{1,2} = \frac{-l \pm \sqrt{l^2 + 4 \frac{g l^2}{2 v_0^2} (h - \frac{g l^2}{2 v_0^2})}}{-2 \frac{g l^2}{2 v_0^2}}$$

za nek l:



za l = max:

$$\tan \varphi_1 = \tan \varphi_2$$

$$\Downarrow$$

$$D = 0$$

$$l^2 + \frac{2 g l^2}{v_0^2} (h - \frac{g l^2}{2 v_0^2}) = 0$$

$$l^2 + \frac{2 g l^2 h}{v_0^2} - \frac{g^2 l^4}{v_0^4} = 0 \quad / : l^2$$

$$1 + \frac{2 g h}{v_0^2} - \frac{g^2 l^2}{v_0^4} = 0$$

$$g^2 l^2 = v_0^4 + 2 g h v_0^2$$

$$l^2 = \frac{v_0^4}{g^2} + \frac{2 h v_0^2}{g}$$

$$l = \sqrt{\frac{v_0^4}{g^2} + \frac{2h v_0^2}{g^2}} = \frac{v_0}{g} \sqrt{(v_0^2 + 2gh)}$$

$$\tan \varphi = \frac{+l}{+\cancel{\frac{g l^2}{2v_0^2}}} = \frac{\frac{v_0}{g} \sqrt{v_0^2 + 2gh}}{\frac{g v_0^2 (v_0^2 + 2gh)}{g v_0^4}} = \frac{v_0 \sqrt{v_0^2 + 2gh}}{v_0^2 + 2gh} = \frac{v_0}{\sqrt{v_0^2 + 2gh}}$$

test:

ce  $h=0$ :  $\varphi = 45^\circ$  ✓

$$\tan \varphi = \frac{10 \frac{m}{s}}{\sqrt{100 + 2 \cdot 10 \cdot 12}} \approx 0,54$$

$$\underline{\underline{\varphi = 28,5^\circ}}$$

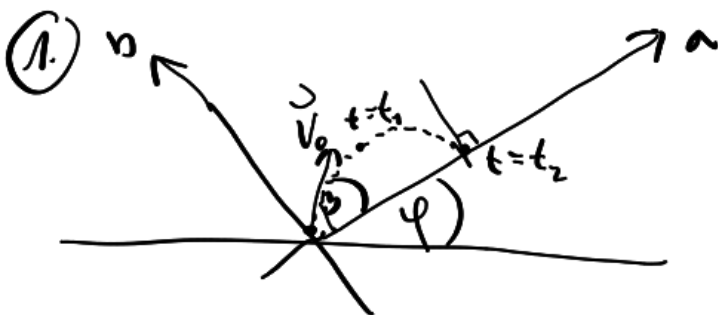
2.3

$\beta = ?$



$\beta_1$  ... po enen odbojih

$\beta_2$  ... po n odbojih



$$t = t_2:$$

$$v_a = 0$$

$$v_b = -v_0$$

$$v_0 \cos \beta = g t_2 \sin \varphi$$

$$v_a = v_0 \cos \beta - g t \sin \varphi$$

$$v_b = v_0 \sin \beta - g t \cos \varphi$$

$$\int v dt = 0 + v_0 t \sin \beta - \frac{g t^2 \cos \varphi}{2}$$

$$t_2 = \frac{v_0 \cos \beta}{g \sin \varphi}$$

$$x_b = t(v_0 \sin \beta - \frac{gt \cos \varphi}{2})$$

$$t = t_1:$$

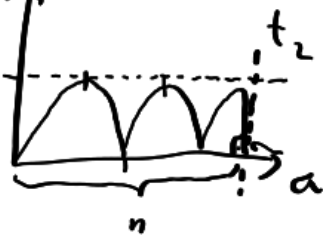
$$v_b = 0 \quad t_1 = \frac{v_0 \sin \beta}{g \cos \varphi}$$

$$t_2 = 2t_1$$

$$\frac{v_0 \cos \beta}{g \sin \varphi} = \frac{2v_0 \sin \beta}{g \cos \varphi}$$

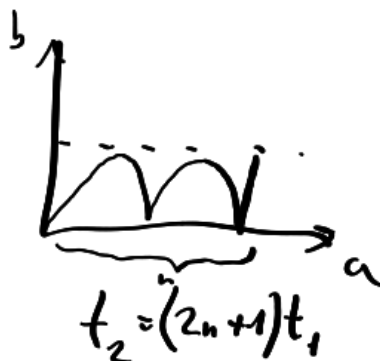
$$\cot \beta_1 = 2 \tan \varphi$$

$\beta_2:$



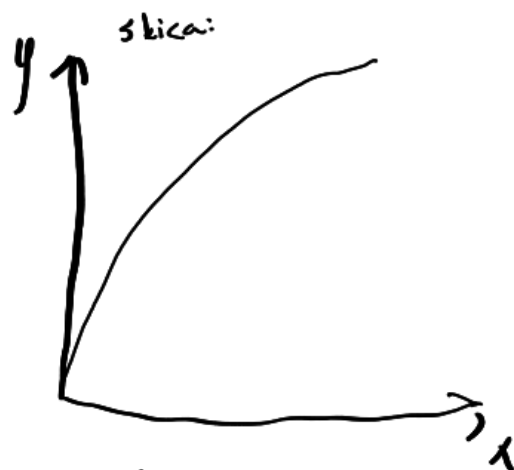
$$t_2 = 2nt_1$$

$$\cot \beta_n = 2n \tan \varphi$$



$$\cot \beta_{2n} = (2n+1) \tan \varphi$$

2.4  $v_y = 3 \text{ m/s}$   
 $v_x = \alpha y$   
 $\alpha = 0,5 \text{ s}^{-1}$   
 $y(x) = ?$



$$y = \int_0^t v_y dt = v_y t$$

$$t = \frac{y}{v_y}$$

$$v_x = \frac{dx}{dt} = \alpha y$$



Hitreje:

$$\alpha_y = \frac{dx}{dt} \cdot \frac{dy}{dy}$$

$$dxV_y = \propto y dy$$

$$xv_y = \alpha \frac{y^2}{2}$$

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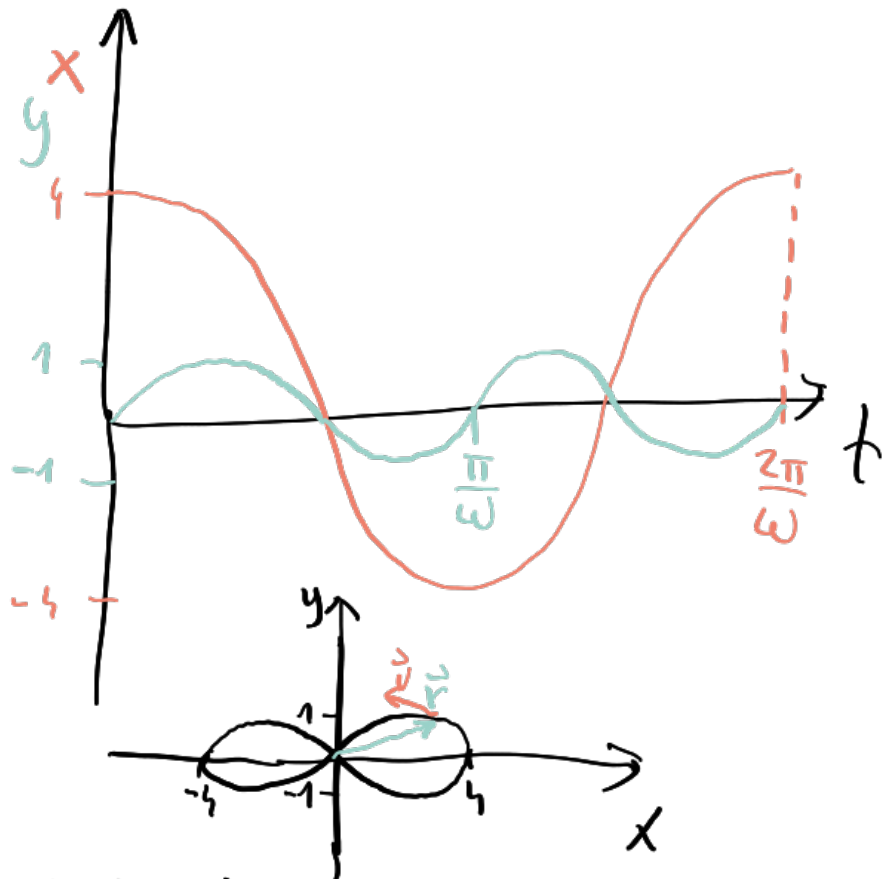
$$x = 4x_0 \cos \omega t$$

$$y = x_0 \sin 2\omega t$$

$$x_0 \approx 10 \text{ cm}$$

$$\underline{\omega = 0,1 \text{ s}^{-1}}$$

$a_{\max} = ?$



$$r = (x, y)$$

$$\vec{v} = \frac{d\vec{r}}{dt} = (-4x_0 \omega \sin \omega t, 2x_0 \omega \cos 2\omega t)$$

$$\vec{a} = \frac{d\vec{v}}{dt} = (-4x_0 \omega^2 \cos \omega t, -4x_0 \omega^2 \sin 2\omega t)$$

$$|\vec{a}| = \sqrt{(-4x_0 \omega^2)^2 (\cos^2 \omega t + \sin^2 2\omega t)}$$

$$|\vec{a}|^2 = 16x_0^2 \omega^4 (\cos^2 \omega t + \sin^2 2\omega t)$$

$$(f(x))' = f'(x)$$

$$(f^2(x))' = 2f(x)f'(x)$$

$$\frac{d|\vec{a}|^2}{dt} = 16x_0^2 \omega^4 ((-2\cos \omega t \cdot \sin \omega t)\omega + (2\sin 2\omega t \cdot 2\omega \cos 2\omega t))$$

$$Q = -\sin 2\omega t + 4 \sin 2\omega t \cos 2\omega t$$

$$Q = \sin 2\omega t (4\cos 2\omega t - 1)$$

$$(1) \sin 2\omega t = 0$$

$$2\omega t = k\pi$$

$$\omega t = \frac{k\pi}{2}$$

$$a^2 = 16x_0^2 \omega^4 (0 + Q) = 0; k = 2n$$

$\Downarrow$   
min



$$a^2 = 16x_0^2 \omega^4; k = 2n$$



$$k \in \mathbb{Z}$$

$$(2) \cos 2\omega t = \frac{1}{4}$$

$$2\omega t = \pm \arccos \frac{1}{4} + 2k\pi$$

$$\omega t = \pm \frac{\arccos \frac{1}{4}}{2} + k\pi$$

$$a^2 = 16x_0^2 \omega^4 (\cos^2(\pm \frac{1}{2} \arccos \frac{1}{4} + k\pi) + \sin^2(\pm \arccos \frac{1}{4} + 2k\pi))$$

$$a^2 = 16x_0^2 \omega^4 (\underbrace{\cos^2(\frac{1}{2} \arccos \frac{1}{4}) + \sin^2(\arccos \frac{1}{4})}_{\geq 1})$$

$\geq 1$

$\Downarrow$

max



1991/92

1. kolokvij (2.)

giba se po elipsi

$$A = 2m$$

$$B = 1m$$

$$t_0 = 0,5s$$

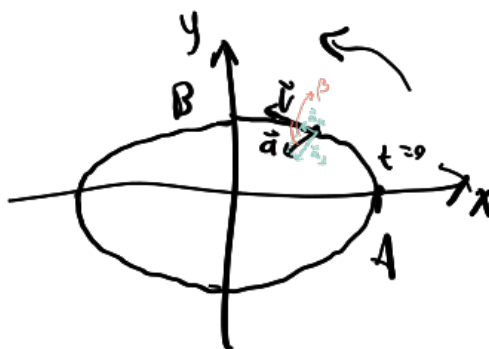
$$x = A \cos(2\pi \frac{t}{t_0})$$

$$y = B \sin(2\pi \frac{t}{t_0})$$

$$a_y = ?$$

$$a_x = ?$$

$$t^* = \frac{t_0}{12}$$



$$\vec{r} = (A \cos(2\pi \frac{t}{t_0}), B \sin(2\pi \frac{t}{t_0}))$$

$$\vec{v} = \frac{d\vec{r}}{dt} = (-A \frac{2\pi}{t_0} \sin(2\pi \frac{t}{t_0}), B \frac{2\pi}{t_0} \cos(2\pi \frac{t}{t_0}))$$

$$\vec{a} = \frac{d\vec{v}}{dt} = (-A \frac{4\pi^2}{t_0^2} \cos(2\pi \frac{t}{t_0}), -B \frac{4\pi^2}{t_0^2} \sin(2\pi \frac{t}{t_0}))$$

$$\vec{a} \cdot \vec{v} = |\vec{a}| \cdot |\vec{v}| \cdot \cos$$

$$\cos \beta = \frac{\vec{a} \cdot \vec{v}}{|\vec{a}| \cdot |\vec{v}|}$$

$$\begin{aligned} \vec{a} \cdot \vec{v} &= A^2 \cdot \frac{8\pi^3}{t_0^3} \cdot \sin \frac{2\pi t}{t_0} \cos \frac{2\pi t}{t_0} - B^2 \frac{8\pi^3}{t_0^3} \sin(2\pi \frac{t}{t_0}) \cos(2\pi \frac{t}{t_0}) = \\ &= A^2 \cdot \frac{8\pi^3}{t_0^3} \cdot \frac{1}{2} \sin(4\pi \frac{t}{t_0}) - B^2 \cdot \frac{8\pi^3}{t_0^3} \cdot \frac{1}{2} \sin(4\pi \frac{t}{t_0}) = \\ &= \frac{4\pi^3}{t_0^3} \sin(4\pi \frac{t}{t_0}) (A^2 - B^2) = 2577 \frac{m^2}{s^3} \end{aligned}$$

$$\begin{aligned} |\vec{v}| &= \sqrt{A^2 \frac{4\pi^2}{t_0^2} \sin^2(2\pi \frac{t}{t_0}) + B^2 \frac{4\pi^2}{t_0^2} \cos^2(2\pi \frac{t}{t_0})} = \\ &= \frac{2\pi}{t_0} \sqrt{A^2 \sin^2(2\pi \frac{t}{t_0}) + B^2 \cos^2(2\pi \frac{t}{t_0})} = 16,6 \frac{m}{s} \\ |\vec{a}| &= \frac{4\pi^2}{t_0^2} \sqrt{A^2 \cos^2(2\pi \frac{t}{t_0}) + B^2 \sin^2(2\pi \frac{t}{t_0})} = 284 \frac{m}{s^2} \end{aligned}$$

$$\cos \beta = \frac{2577}{16,6 \cdot 284} = 0,54$$

$$\beta = 57,2^\circ$$

$$|\vec{a}_y| = |\vec{a}| \cos \beta = 155 \frac{m}{s^2}$$

$$|\vec{a}_x| = |\vec{a}| \sin \beta = 239 \frac{m}{s^2}$$

2.6

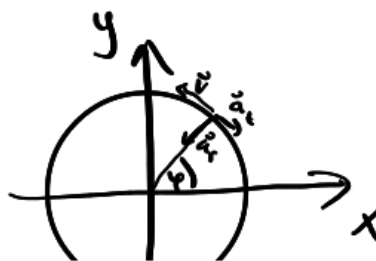
$$R = 1m$$

$$\omega = \frac{v}{R}$$

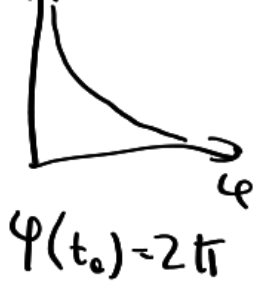
$$a_r = \omega^2 R$$

$$a_t = \alpha R$$

$$\omega =$$



$$\frac{k=1\text{ s}^{-1}}{a_r, a_t=?} t=t_0$$



$$4 \vec{a}, \vec{v}$$

$$a_r = \omega^2 R = \frac{k^2}{2\pi} R = 0,16 \frac{\text{m}}{\text{s}^2}$$

$$\alpha = \frac{d\omega}{dt} \cdot \frac{d\varphi}{d\varphi} = \omega \cdot \frac{d\omega}{d\varphi} =$$

$$= -\frac{1}{2} \omega \cdot k \cdot \varphi^{-\frac{3}{2}}$$

$$a_t = \alpha R = -\frac{1}{2} \frac{k^2}{\sqrt{\varphi} \cdot \sqrt{\varphi}} = -\frac{1}{2} \frac{k^2}{\varphi^2} = -\frac{1}{6\pi^2} \frac{\text{m}}{\text{s}^2}$$

$$= -0,013 \frac{\text{m}}{\text{s}^2}$$

$$\frac{\pi}{2} \dots 90$$

$$1,65 \dots x$$

$$\tan \beta = \frac{|\vec{a}_t|}{|\vec{a}_r|} = \frac{1 \cdot 2\pi}{6\pi^2 \cdot 1} = \frac{1}{3\pi}$$

$$\beta = 0,08$$

$$\chi = \frac{\pi}{2} + \beta = \underline{1,65} = 94,6^\circ$$

29/11

$$\varphi = 60^\circ$$

$$v_0 = 20 \frac{\text{m}}{\text{s}}$$

$$t = 3\text{ s}$$

$$a_t, a_r = ?$$

$$r = ?$$



$$a_r = g \cos \beta$$

$$a_t = g \sin \beta$$

$$v_x = v_0 \cos \varphi$$

$$v_y = v_0 \sin \varphi - g t$$

$$t = 3\text{ s}:$$

$$v_x = 10 \frac{\text{m}}{\text{s}}$$

$$v_y = -12,7 \frac{\text{m}}{\text{s}}$$

$$\frac{|v_y|}{|v_x|} = \tan \beta$$

$$\beta = 0,90 = 52^\circ$$

$$a_r = 6,2 \frac{\text{m}}{\text{s}^2}$$

$$a_t = 7,8 \frac{\text{m}}{\text{s}^2}$$

$$a_r = \omega^2 r = \frac{v^2}{r}$$

$$r = \frac{v^2}{a_r} = \frac{v_x^2 + v_y^2}{a_r} = \underline{42\text{ m}}$$

29/11

$$\alpha = \frac{d\omega}{dt} \cdot \frac{d\varphi}{d\varphi} = \omega \cdot \frac{d\omega}{d\varphi}$$

$$\omega_0 = 6 \text{ s}^{-1}$$

$$\alpha < 0, \text{ konst.}$$

$$\frac{t_{10} \dots \omega_{10} = 0}{t_5 - t_1 = ?} \quad \varphi = 20\pi$$

$$\int_{\varphi_0}^{\varphi} d\varphi = \int_{\omega_0}^{\omega} \frac{1}{\alpha} d\omega$$

$$\alpha \varphi = \frac{\omega^2 - \omega_0^2}{2}$$

$$\omega = \sqrt{\omega_0^2 + 2\alpha\varphi}$$

$$0 = \omega_0^2 + 2 \cdot \alpha \cdot 20\pi$$

$$\alpha = -\frac{\omega_0^2}{40\pi} = -0,29/\text{s}^2$$

$$t_0 = \frac{\omega_N - \omega_0}{\alpha} = \frac{\sqrt{\omega_0^2 + 2\alpha\varphi} - \omega_0}{\alpha}$$

$$t_5 = 6,15 \text{ s}$$

$$t_1 = 4,73 \text{ s}$$

$$t_5 - t_1 = \underline{\underline{1,42 \text{ s}}}$$

9/19

Suzno nihalo

$$\nu = 0,5 \text{ Hz}$$

$$\varphi_0 = 30^\circ$$

$$a = ?$$

$$b = ?$$

$$r = 10 \text{ cm}$$

$$\varphi = 10^\circ$$



$$a_t = \alpha r$$

$$a_r = \omega^2 r$$

$$\varphi = \varphi_0 \sin \omega_0 t$$

$$\omega = \frac{d\varphi}{dt} = \varphi_0 \omega_0 \cos \omega_0 t$$

$$\alpha = \frac{d\omega}{dt} = -\varphi_0 \omega_0^2 \sin \omega_0 t$$

$$\omega_0 = 2\pi\nu$$

$$a_t = -\varphi_0 (2\pi\nu)^2 \cdot r \sin 2\pi\nu t$$

$$a_r = \omega^2 r = \varphi_0^2 (2\pi\nu)^2 r \cos^2 2\pi\nu t$$

$$a^2 = a_t^2 + a_r^2 = \varphi_0^2 r^2 (2\pi\nu)^4 (\sin^2(2\pi\nu t) + \cos^2(2\pi\nu t))$$

$$\varphi_1 = \varphi_0 \sin \omega_0 t_1$$

$$\sin \omega_0 t_1 = \frac{1}{3}$$

$$\omega_0 t_1 = \arcsin \frac{1}{3}$$

$$\omega_0 t_1 = 0,3398$$

$$a^2 = \varphi_0^2 r^2 (2\pi\nu)^4 \left( \sin^2(0,3398) + \varphi_0^2 \cos^2(0,3398) \right) = 0,0875 \left( \frac{\text{m}}{\text{s}^4} \right)^2$$

$$a = 0,30 \frac{\text{m}}{\text{s}^2}$$

$$\tan \beta = \frac{a_t}{a_r} = \frac{\sin^2(0,3398)}{\cos^2(0,3398)} = 0,1406$$

$$\underline{\underline{p = 0,14}}$$

29/2

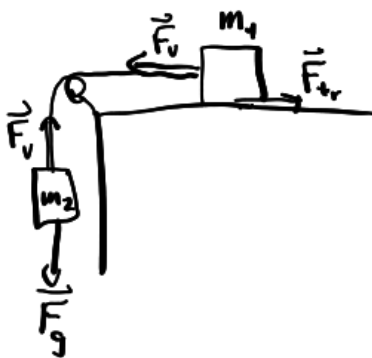
$$m_1 = 1,2 \text{ kg}$$

$$m_2 = 0,4 \text{ kg}$$

$$x_1 = 1 \text{ m}$$

$$t_1 = 1,1 \text{ s}$$

$$F_{tr} = ?$$



$$m_1 a = F_v - F_{tr}$$

$$m_2 a = F_g - F_v$$

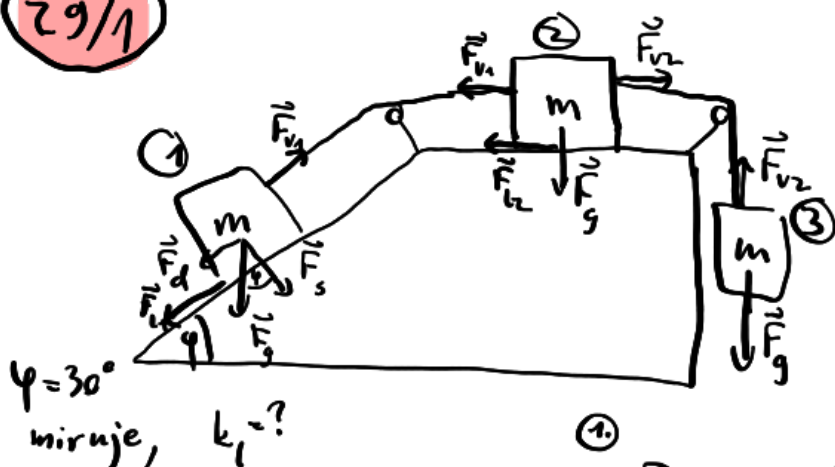
$$(m_1 + m_2) a = m_2 g - F_{tr}$$

$$x = \frac{a t^2}{2}$$

$$a = \frac{2x}{t^2} = 1,65 \frac{\text{m}}{\text{s}^2}$$

$$F_{tr} = m_2 g - (m_1 + m_2) a = \underline{1,36 \text{ N}}$$

29/1



$\varphi = 30^\circ$   
miruje,  $k_l = ?$

①

$$F_{v1} = F_d + F_{L1}$$

$$F_d = mg \sin \varphi$$

②

$$F_{v2} = F_{v1} + F_{L2} \rightarrow F_{v1} = F_g - F_{L2}$$

$$F = F_{L1} = mg \cos \varphi$$

$$F = F_{L2}$$

$$F_{L1} = F_g k_L = mg \cos \varphi$$

$$F_{L2} = F_g k_L = mg k_L$$

$$F_g - F_{L2} = F_d + F_{L1}$$

$$mg - mg k_L = mg \sin \varphi + mg \cos \varphi k_L$$

$$1 - k_L = \sin \varphi + \cos \varphi k_L$$

$$k_L (1 + \cos \varphi) = 1 - \sin \varphi$$

$$k_L = \frac{1 - \sin \varphi}{1 + \cos \varphi} = \underline{\underline{0,27}}$$

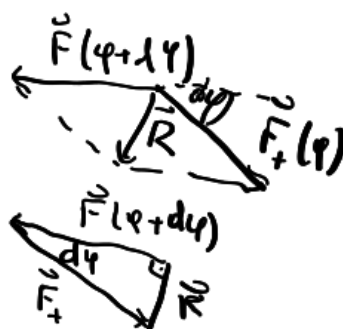
29/16

$$r = 10 \text{ cm}$$

$$F_1 = 2000 \text{ N}$$

$$k_{tr} = 0,6$$

$$F_2 = ?$$



$$R = F \sin d\varphi \approx F(\varphi) d\varphi$$

$$dF_{tr} = dR k_t = d\varphi \cdot F(\varphi) \cdot k_t$$

$$F(\varphi + d\varphi) = F(\varphi) + dF$$

$$\int_{F_2}^F \frac{dF}{F} = \int_0^\varphi k_t d\varphi$$

$$\ln F \Big|_{F_2}^F = k_t \varphi \Big|_0^\varphi$$

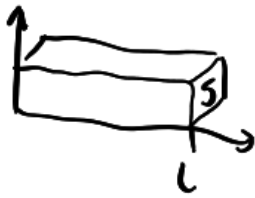
$$\ln \frac{F}{F_2} = k_t \varphi$$

$$F = F_2 \cdot e^{k_t \varphi}$$

$$F_2 = F_1 e^{-k_t \varphi} =$$

$$= 2 \text{ kN} \cdot e^{-0,6 \cdot \pi} = \underline{\underline{0,3 \text{ kN}}}$$

težišće



$$\rho = \frac{x}{l} \rho_0$$

$$m_{x_t} = \int_0^l x dm = \int_0^l x \cdot \rho dV = \int_0^l x \rho S dx = \frac{\rho_0 S}{l} \int_0^l x^2 dx$$

$$x_t = \frac{\rho_0 S}{l m} \cdot \frac{l^3}{3}$$

$$m = \int_0^l dm = \int_0^l \frac{x}{l} \rho_0 S dx = \frac{\rho_0 S}{l} \frac{l^2}{2} = \frac{\rho_0 S l}{2}$$

$$x_t = \frac{\rho_0 S l \cdot 2}{3 \cdot \rho_0 S l} = \frac{2}{3} l$$

3.6

$$m$$

$$k_1 = k_2$$

$$F_m, F_M = ?$$

$$a_n \text{ je } F > F_m = ?$$

$$a_n \text{ je } F < F_m = ?$$



$$mg = \vec{F}_1 + \vec{F}_2$$

$$mg = F_1 + F_1 e^{-k_1 \pi}$$

$$\frac{mg}{1 + e^{-k_1 \pi}} = F_m$$

$$\frac{mg}{1 + e^{k_1 \pi}} = F_m$$

$$F_1 + F_2 - mg = ma$$

$$a = \frac{F_1 + F_1 e^{-k_1 \pi} - mg}{m} = \frac{\left( \frac{mg}{1 + e^{-k_1 \pi}} + \Delta F \right) (1 + e^{-k_1 \pi}) - mg}{m} =$$

$$= \frac{mg + \Delta F (1 + e^{-k_1 \pi}) - mg}{m} = \frac{\Delta F (1 + e^{-k_1 \pi})}{m}$$



Kolokvij

1990/91 / 2. nal

$$F = 500 \text{ N}$$

$$m_0 = 30 \text{ kg}$$



na dlekuca (d):

$$-m_d g + F + F_v = m_d a_d$$

na odraz:

$$m_d = 90 \text{ kg}$$

$$a = ?$$

$$F_s = ?$$

škripec:

$$F_s - 2F_v = 0$$

$$F_s = 2F_v$$

$$F_v = m_o a + m_o g + F =$$

$$= \frac{30 \cdot 20 \cdot 10}{3 \cdot 1} + 300 + 500 =$$

$$= 1000 \text{ N}$$

$$F_s = 2 \text{ kN}$$



$$-m_o g - F + F_v = m_o a_o$$

$$a_o = a_d$$

$$-m_d g + F + m_o g + F = a(m_d - m_o)$$

$$\frac{2F + g(m_o - m_d)}{m_d - m_o} = a$$

$$\frac{1000 \text{ N} + g(-60 \text{ kg})}{60 \text{ kg}} = a$$

$$a = \frac{400}{60} \frac{\text{m}}{\text{s}^2} = 6,67 \frac{\text{m}}{\text{s}^2}$$

3.8

$(, x_o$   
zan. trenje

$$V(x) = ?$$

$$x(t) = ?$$

$$t_k = ?$$



$$m_1 = \frac{l-x}{l} m$$

$$m_2 = \frac{x}{l} m$$

$$m_2 g = (m_1 + m_2) a$$

$$a = \frac{\frac{x}{l} m g}{m(\frac{l-x}{l} + \frac{x}{l})} = \frac{x}{l} g$$

$$a = \frac{dv}{dt} \cdot \frac{dx}{dx} = \frac{dv}{dx} v$$

$$v dv = \frac{x}{l} g dx$$

$$\int_0^v v dv = \frac{g}{l} \int_{x_o}^x x dx$$

$$\frac{v^2}{2} = \frac{g}{l} \left( \frac{x^2 - x_o^2}{2} \right)$$

$$v = \sqrt{\frac{g}{l} (x^2 - x_o^2)}$$

$$v = \frac{dx}{dt} = \sqrt{\frac{g}{l} (x^2 - x_o^2)}$$

$$\int_0^t dt = \int_{x_o}^x \frac{1}{\sqrt{\frac{g}{l} (x^2 - x_o^2)}} dx$$

$$x = x_o \operatorname{ch}(u)$$

$$dx = x_o \operatorname{sh}(u) du$$

$$t = \int_{\operatorname{arccch}(1)}^{\operatorname{arccch} \frac{x}{x_o}} \frac{x_o \operatorname{sh}(u) du}{\sqrt{\frac{g}{l} (x_o^2 \operatorname{ch}^2(u) - x_o^2)}} = \int_0^{\operatorname{arccch} \frac{x}{x_o}} \frac{x_o \operatorname{sh}(u) du}{x_o (\sqrt{\frac{g}{l}} \operatorname{sh}(u))} = \sqrt{\frac{l}{g}} \operatorname{arccch} \frac{x}{x_o}$$

Reševanje ...

$$\ddot{x} = \frac{g}{l} x \quad \text{zač. pog.: } v_0 = 0 \\ x_0 = x(t=0)$$

$$x = A e^{\frac{t}{\tau}} + B e^{-\frac{t}{\tau}}$$

$$\dot{x} = A \frac{1}{\tau} e^{\frac{t}{\tau}} - B \frac{1}{\tau} e^{-\frac{t}{\tau}} \rightarrow 0 = A \sqrt{\frac{g}{l}} \cdot 1 - B \sqrt{\frac{g}{l}} \cdot 1$$

$$\ddot{x} = A \frac{1}{\tau^2} e^{\frac{t}{\tau}} + B \frac{1}{\tau^2} e^{-\frac{t}{\tau}}$$

$$A = B$$

$$x_0 = A + B = 2A$$

$$A = \frac{x_0}{2} = B$$

$$= \frac{1}{\tau^2} (A e^{\frac{t}{\tau}} + B e^{-\frac{t}{\tau}}) =$$

$$= \frac{1}{\tau^2} x$$

$$\frac{g}{l} x = \frac{1}{\tau^2} x$$

$$\tau = \sqrt{\frac{l}{g}}$$

$$x(t) = x_0 \operatorname{ch}(t \sqrt{\frac{g}{l}})$$

$$\frac{x}{x_0} = \operatorname{ch}(t \sqrt{\frac{g}{l}})$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2i}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\operatorname{ch}(x) = \frac{1}{2} (e^x + e^{-x})$$

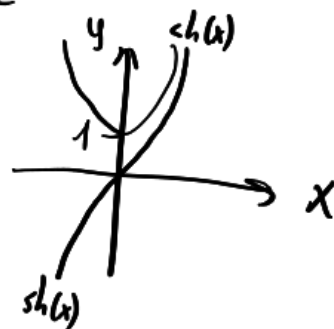
$$\operatorname{sh}(x) = \frac{1}{2} (e^x - e^{-x})$$

$$\operatorname{ch}^2 - \operatorname{sh}^2 = 1$$

$$\frac{d \operatorname{ch}(x)}{dx} = \operatorname{sh}(x)$$

$$\frac{d \operatorname{sh}(x)}{dx} = \operatorname{ch}(x)$$

hiperbolični kosinus



$$t = \sqrt{\frac{l}{g}} \operatorname{arccch} \frac{x}{x_0}$$

$$t_k = \sqrt{\frac{l}{g}} \operatorname{arccch} \frac{l}{x_0}$$

3.11

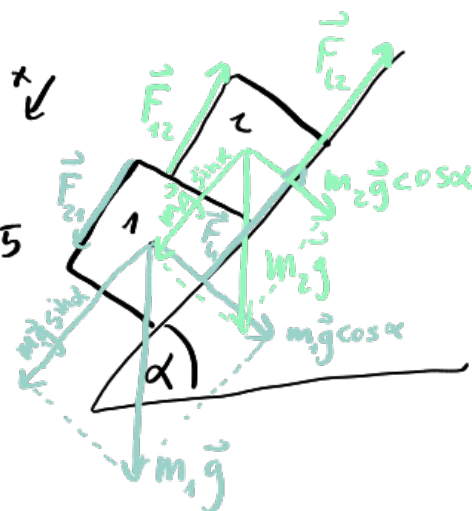
$$m_1 = 3 \text{ kg}$$

$$m_2 = 2 \text{ kg}$$

$$k_{t1} = 0,2 \quad k_{t1} = 0,25$$

$$k_{t2} = 0,05 \quad k_{t2} = 0,1$$

$$\alpha_2 = ? \quad \text{z drs}$$



$$\vec{F}_{12} = -\vec{F}_{21}$$

$$\alpha = 30^\circ$$

$$F_{12} = ?$$

$$m_2 g \sin \alpha = F_{12} + F_{t2}$$

$$\therefore F_{12} + m_1 g \sin \alpha = F_{t1}$$

$$m_2 g \sin \alpha = k_{t2} m_2 g \cos \alpha + k_{t1} m_1 g \cos \alpha - m_1 g \sin \alpha$$

$$\sin \alpha (m_1 + m_2) = \cos \alpha (k_{t1} m_1 + k_{t2} m_2)$$

$$\tan \alpha = \frac{m_1 + m_2}{k_{t1} m_1 + k_{t2} m_2}$$

$$\alpha = 11^\circ$$

$$\alpha = 30^\circ$$

$$1: F_{12} + m_1 g \sin \alpha - F_{t1} = m_1 a$$

$$2: m_2 g \sin \alpha - F_{12} - F_{t2} = m_2 a$$

$$\frac{F_{12}}{m_1} + g \sin \alpha - k_{t1} g \cos \alpha = g \sin \alpha - \frac{F_{12}}{m_2} - k_{t2} g \cos \alpha$$

$$F_{12} \left( \frac{1}{m_1} + \frac{1}{m_2} \right) = g \cos \alpha (k_{t1} - k_{t2})$$

$$F_{12} = \frac{10 \frac{\sqrt{3}}{2} (0,2 - 0,05) \frac{\text{m}}{\text{s}^2} \cdot k g}{\frac{1}{2} + \frac{1}{3}} = \underline{\underline{1,6 \text{ N}}}$$

$$200 \text{ g/1e} \quad 1. \text{kol/4}$$

$$m = 2 \text{ kg}$$

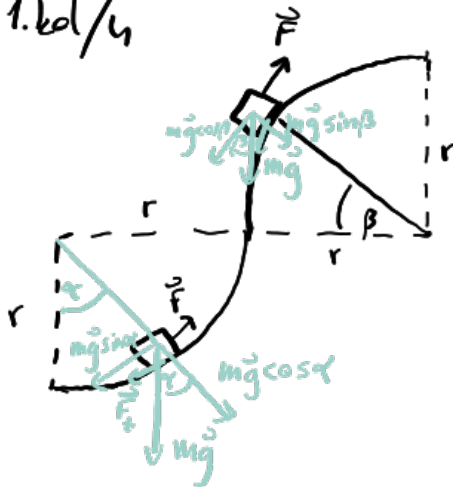
$$r = 20 \text{ m}$$

$$k_t = 0,3$$

$$a = 0$$

$$F \dots \text{vzp. skl.}$$

$$A = ?$$



$$F = m g \sin \alpha + k_t m g \cos \alpha = m g (\sin \alpha + k_t \cos \alpha)$$

$$A_1 = \int m g (\sin \alpha + k_t \cos \alpha) ds$$

$$ds = r d\alpha$$

$$A_1 = m g r \int_0^{\frac{\pi}{2}} (\sin \alpha + k_t \cos \alpha) d\alpha =$$

$$= m g r (k_t \sin \alpha - \cos \alpha) \Big|_0^{\frac{\pi}{2}} =$$

$$= m g r (k_t - 0 - 0 + 1) = 400 \cdot 1,3 \text{ J}$$

$$F = k_t m g \sin \beta + m g \cos \beta$$

$$A_2 = \int_0^{\frac{\pi}{2}} m g r (k_t \sin \beta + \cos \beta) d\beta$$

$$(k_t \cos \beta + \sin \beta) \Big|_0^{\frac{\pi}{2}} =$$

$$A_2 = mgr(-k_t \cos \varphi + \sin \varphi)$$

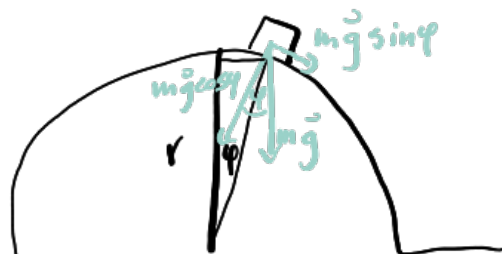
$$= mgr(0 + 1 + k_t - 0) = 400 \cdot 1,3 \text{ J}$$

$$A = 800 \cdot 1,3 \text{ J} = 1040 \text{ J} = 1 \text{ kJ}$$

3.13

$k_{tr=0}$

kadaj se odlepi?



$$\Delta W_p = \Delta W_k$$

$$mgh = \frac{1}{2}mv^2$$

$$\Delta h = r - r \cos \varphi = r(1 - \cos \varphi)$$

$$v = \sqrt{2gr(1 - \cos \varphi)}$$

$$a_r = \frac{v^2}{r} = 2g(1 - \cos \varphi)$$

odlepi se, ko:

$$g \cos \varphi = 2g(1 - \cos \varphi)$$

$$3 \cos \varphi = 2$$

$$\cos \varphi = \frac{2}{3}$$

$$\varphi = 48^\circ$$

3.10

$$a = 0,7 \frac{m}{s^2}$$

$$l = 1m$$

$$\varphi_0 = ?$$



$$\vec{a}_s = -\vec{a}_p$$

$$\frac{\vec{a}_s}{g} = \tan \varphi_0$$

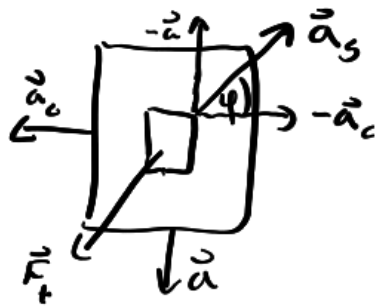
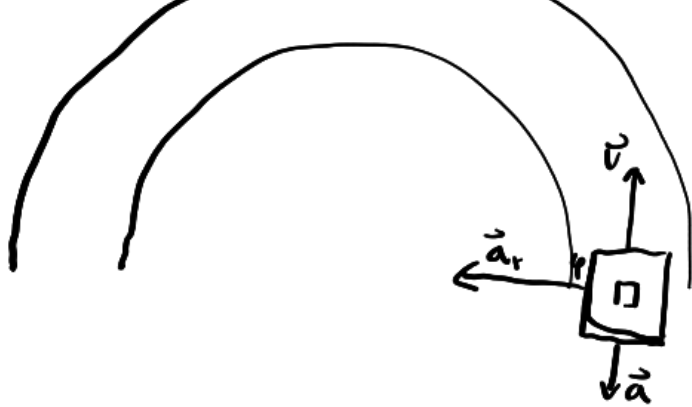
$$\varphi_0 = 0,07 = 4^\circ$$

$$a = \frac{g}{2}$$

$$r = 20\text{m}$$

$$k_t = 0,6$$

$$v_n = ?$$



$$\frac{v^2}{r} \cdot m = F_t \cos \alpha$$

$$\vec{a} \cdot m = F_t \sin \alpha$$

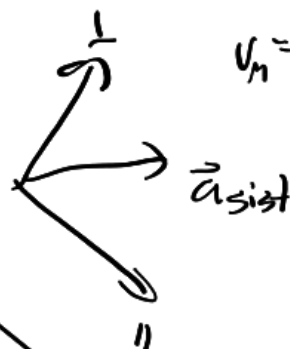
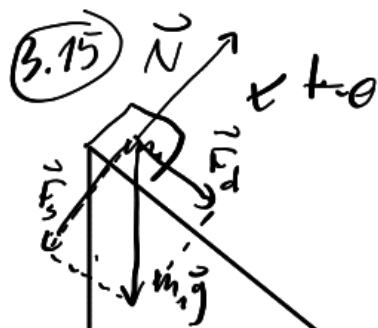
$$F_t = mg k_{tr}$$

$$\frac{v_r^2}{r} = k_t g \cos \alpha$$

$$\vec{a} = k_t g \sin \alpha$$

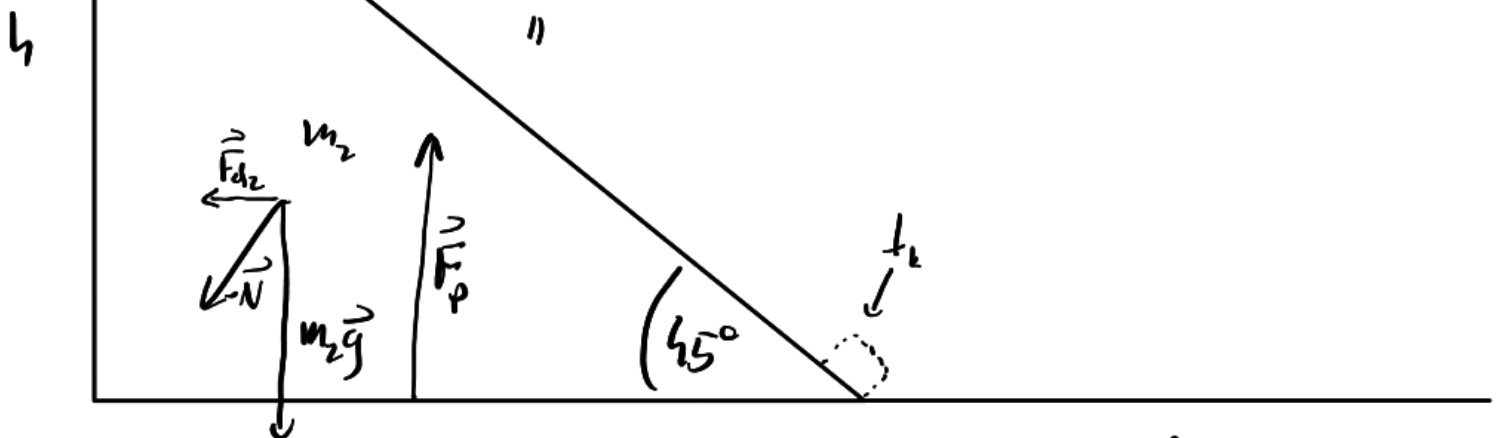
$$a_s^2 = a_c^2 + a^2 = \frac{v_n^4}{r^2} + a^2$$

$$\left(\frac{F_t}{m}\right)^2 = (g k_{tr})^2$$



$$v_n^4 = g^2 k_{tr}^2 r^2 + a^2 r^2$$

$$v_n = \sqrt{\dots} = 8,14 \frac{\text{m}}{\text{s}}$$



$$m_2, m_1$$

$$k_{tr} = 0$$

$$t_1 = ?$$

$$\vec{x}$$

$$2. x:$$

$$-F_{d2} = m_2 a_2$$

$$\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

$$x_2 = ?$$

$$m_1 \gg m_2$$

$$m_2 \gg m_1$$

$$m_2 a_2 = -N \cos \varphi \rightarrow a_2 = -\frac{N\sqrt{2}}{2m_2}$$

1.

$$a_{\text{sis} \parallel} = -a_2$$

$$\parallel: F_d + m_1 a_{\text{sis} \parallel} = m_1 a_{\parallel}$$

$$g \frac{\sqrt{2}}{2} + a_{\text{sis} \parallel} = a_{\parallel}$$

$$a_{\parallel} = \frac{\sqrt{2}}{2} \left( g + \frac{N}{m_2} \right)$$

$$\perp: -F_s + N + F_{\text{sis} \perp} = 0$$

$$-m_1 g \frac{\sqrt{2}}{2} + N + m_1 a_{\text{sis} \perp} = 0$$

$$N + m_1 a_2 \frac{\sqrt{2}}{2} = m_1 g \frac{\sqrt{2}}{2}$$

$$N \left( 1 + \frac{m_1}{2m_2} \right) = N + m_1 \frac{N\sqrt{2}}{2m_2} \cdot \frac{\sqrt{2}}{2} = m_1 g \frac{\sqrt{2}}{2}$$

$$N = \frac{m_1 g}{\left( 1 + \frac{m_1}{2m_2} \right) \sqrt{2}}$$

$$\begin{aligned} a_{\parallel} &= \frac{\sqrt{2}}{2} \left( g + \frac{N}{m_2} \right) = \frac{\sqrt{2}}{2} \left( g + \frac{m_1 g}{\left( 1 + \frac{m_1}{2m_2} \right) \sqrt{2} m_2} \right) = \\ &= \frac{\sqrt{2}}{2} g \left( 1 + \frac{m_1}{\left( m_2 + \frac{m_1}{2} \right) \sqrt{2}} \right) = \frac{1}{2} g \left( \sqrt{2} + \frac{m_1}{m_2 + \frac{m_1}{2}} \right) \end{aligned}$$

$$h\sqrt{2} = \frac{a_{\parallel} t_k^2}{2}$$

$$2h\sqrt{2} = t_k^2 \left( \frac{1}{2} g \left( \sqrt{2} + \frac{m_1}{m_2 + \frac{m_1}{2}} \right) \right)$$

$$t_k = \sqrt{\frac{4h\sqrt{2}}{g \left( \sqrt{2} + \frac{m_1}{m_2 + \frac{m_1}{2}} \right)}}$$

$$a_2 = -\frac{m_1 g}{2 \left( 1 + \frac{1}{2} \frac{m_1}{m_2} \right) m_2} = -\frac{m_1 g}{2 \left( m_2 + \frac{1}{2} m_1 \right)}$$

$$x_2 = \frac{1}{2} a_2 t_k^2 = - \frac{1}{2} \frac{m_1 g \cdot 4h\sqrt{2}}{(m_2 + \frac{1}{2}m_1)g(\sqrt{2} + \frac{m_1}{m_2+m_1})} = - \frac{m_1 h \sqrt{2}}{(m_2 + \frac{1}{2}m_1)(\sqrt{2} + \frac{m_1}{m_2+m_1})}$$

29/10

$\nu = 4 \text{ Hz}$   
 odlik -  $x_{\text{on}} = ?$

- če  $x_0$  majhen, utež nika skupaj  $\rightarrow$  ploščice
- če  $x_0$  velik, utež poskakuje
- $x_{\text{on}}$  - meja med poskakanjem



$$x = x_0 \sin \omega t = x_0 \sin 2\pi \nu t$$

$$\ddot{x} = -x_0 \omega^2 \sin \omega t = a_p = -\omega^2 x$$

$$\underbrace{-a_p}_{a_s} m - mg = ma$$

$$\omega^2 x - g = a \stackrel{\text{na meji}}{=} 0$$

$$\omega^2 x = g$$

$$x = \frac{g}{\omega^2}$$



$\nu$  zg. legi

$$x_{\text{on}} = \frac{g}{\omega^2} = \frac{g}{4\pi^2 \nu^2} = \frac{10}{64\pi^2} \text{ m} = \underline{\underline{1,6 \text{ cm}}}$$

29/24

$$m_i = 100 \text{ g}$$

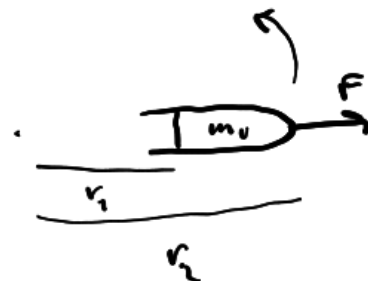
$$\nu = 120 \text{ Hz}$$

$$r_1 = 8 \text{ cm}$$

$$r_2 = 15 \text{ cm}$$

$$F = ?$$

Floris:



$$a_r = \omega^2 r$$

$$dF = dm \cdot a_r = dm \cdot \omega^2 r = \omega^2 \cdot \frac{m}{r_2 - r_1} \cdot r dr$$

$$\int_{r_1}^{r_2} \dots = 2m \frac{r_2^2 - r_1^2}{2} = \dots$$

$$F = \int_{r_1}^{r_2} \omega^2 r_2 r_1 r dr = \omega^2 \frac{r_2^3 - r_1^3}{3}$$

$$= (2\pi \nu)^2 \frac{m}{r_2 - r_1} \frac{r_2^3 - r_1^3}{3} = 2(\pi \nu)^2 m(r_2 + r_1) \approx$$

$$\approx \underline{\underline{6,5 \cdot 10^3 \text{ N}}}$$

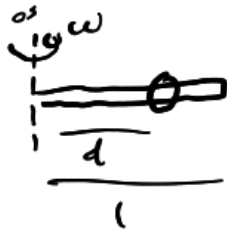
29/23

$$l = 1 \text{ m}$$

$$d = 0,7 \text{ m}$$

$$\omega = 0,7 \text{ s}^{-1}$$

$$v_r = ?$$



$$a_c = \omega^2 r = \frac{dv_r}{dt} \cdot \frac{dr}{dr} = v_r \cdot \frac{dv_r}{dr}$$

$$\int_d^l \omega^2 r dr = \int_{v_r}^{v_l} v_r dv_r$$

$$\frac{v_r^2}{2} = \omega^2 \frac{r^2}{2} \Big|_d^l = \omega^2 (l^2 - d^2) = \omega^2 (0,51 \text{ m}^2)$$

$$v_r = 0,50 \frac{\text{m}}{\text{s}}$$

$$a = -2\vec{\omega} \times \vec{v} - \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$a_{cor} = -2\vec{\omega} \times \vec{v}$$

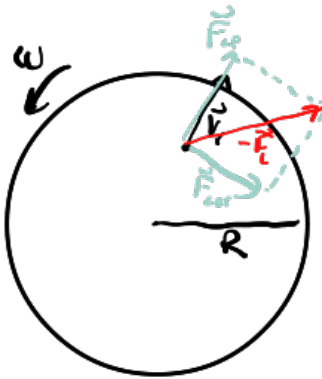
3.19

$$R = 2 \text{ m}$$

$$\omega = 0,5 \text{ s}^{-1}$$

$$v_r = 1 \frac{\text{m}}{\text{s}}$$

$$k_l = ?$$



$$\vec{F}_l = \vec{F}_{cor} + \vec{F}_{ce}$$

$$F_l^2 = F_{cor}^2 + F_{ce}^2$$

$$(\mu g k_l)^2 = (\mu 2 v_r \omega)^2 + (\mu \omega^2 r)^2$$

$$k_l = \frac{1}{g} \sqrt{4 v_r^2 \omega^2 + \omega^2 r^2} = \frac{\omega}{g} \sqrt{4 v_r^2 + \omega^2 r^2}$$

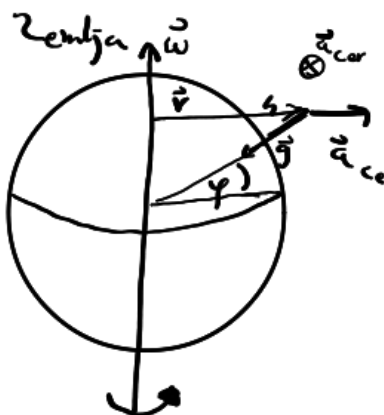
$$\approx 0,11$$

29/19

$$\varphi = 30^\circ$$

$$h = 125 \text{ m}$$

$$x = ?$$



$$\vec{a} = -2\vec{\omega} \times \vec{v} - \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$a_{ce} = \omega^2 r = \left(2\pi \frac{1}{24 \cdot 60^2}\right)^2 \cdot (6400 \cdot 125 \text{ m}) \cdot \cos 30^\circ$$

$$r = (R + h) \cos \varphi$$

premik proti V

$$a_{ce} = 0,03 \frac{m}{s^2}$$

⊗ v table

⊙ iz table



$$a_{ce} \ll g \Rightarrow \vec{g} \parallel \vec{v}$$

$$a_{cor} = 2\omega v \sin(90 + \varphi) = 2\omega v \cos \varphi$$

$$\underbrace{v = gt}_?$$

$$a_{cor} = 2\omega g t \cos \varphi$$

$$v_x = \int_0^t a_{cor} dt = \omega g t^2 \cos \varphi$$

$$x = \int_0^t v_x dt = \omega g \frac{t^3}{3} \cos \varphi$$

$$x = \frac{1}{3} \omega g \left( \frac{2h}{g} \right)^{\frac{3}{2}} \cos \varphi = 2,6 \text{ cm}$$

$$h = \frac{g t_f^2}{2}$$

$$t_f = \sqrt{\frac{2h}{g}}$$

3.23

$$l = 10 \text{ m}$$

$$k = 80 \frac{N}{m}$$

$$m = 80 \text{ kg}$$

$$v_{max} = ?$$

$$h(v_{max}) = ?$$

$$h_{max} = ?$$

$$F_g - F_{el} = ma$$

$$v_{max}, \text{ ko } a = 0 :$$

$$mg = kx$$

$$x = \frac{800 \cancel{N} \text{ m}}{80 \cancel{N}} = 10 \text{ m}$$

$$h(v_{max}) = l + x = 20 \text{ m}$$

$$-mgh + \frac{mv^2}{2} + \frac{kx^2}{2} = 0$$

$$v^2 = \frac{2mgh - kx^2}{m} = \frac{2 \cdot 800 \cdot 20 - 8000}{80} = 40 \cdot 10 - 100 = \underline{\underline{300}}$$

$$v = 17,3 \frac{m}{s}$$

$$h_{\max}: v=0$$

$$kx^2 = 2mg(x+l)$$

$$kx^2 - 2mgx - 2mgl = 0$$

$$80x^2 - 2 \cdot 800x - 2 \cdot 800 \cdot 10 = 0$$

$$x^2 - 2 \cdot 10x - 2 \cdot 10 \cdot 10 = 0$$

$$x^2 - 20x - 200 = 0$$

$$x_{1,2} = \frac{20 \pm \sqrt{400 + 800}}{2} = 10 \pm 10\sqrt{3}$$

$$x_{\max} = 10 + 10\sqrt{3}$$

$$h_{\max} = 20 + 10\sqrt{3} \text{ m} = \underline{37 \text{ m}}$$

$$2g/2$$

$$m = 500 \text{ kg}$$

$$F = 5200 \text{ N}$$

$$P(10s) = ?$$

$$A = \vec{F} \cdot \vec{s}$$

$$\neq$$
  

$$\text{ker } F \text{ konst.}$$

$$P = \vec{F} \frac{d\vec{s}}{dt} = \vec{F} \vec{v}$$

$$F - mg = ma$$

$$a = 0,4 \text{ m/s}^2$$

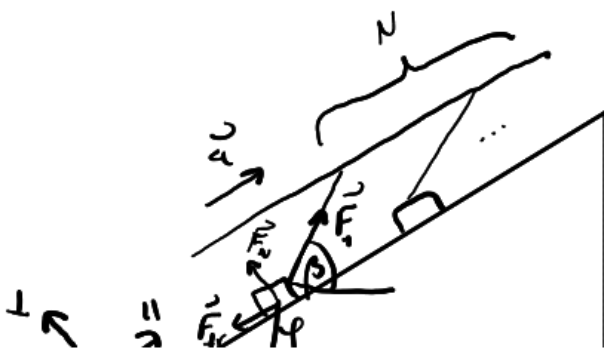
$$v = at = 4 \frac{\text{m}}{\text{s}}$$

$$P = Fv = 4 \cdot 5200 \text{ N} \frac{\text{m}}{\text{s}} = 20800 \text{ W} = 21 \text{ kW}$$

$$N = 50$$

$$a = 1 \frac{\text{m}}{\text{s}^2}$$

$$P(1s) = ?$$



$$\vec{v} = \vec{a}t$$

$$P(2s) = ?$$

$$k_t = 0,1$$

$$\varphi = 30^\circ$$

$$\beta = 70^\circ$$

$$m = 80 \text{ kg}$$

$$\downarrow: F_N + F_1 \sin(\beta - \varphi) - mg \cos \varphi = 0$$

$$\parallel: F_1 \cos(\beta - \varphi) - mg \sin \varphi - k_t F_N = ma$$

$$F_1 \cos(\beta - \varphi) - mg \sin \varphi - k_t (mg \cos \varphi - F_1 \sin(\beta - \varphi)) = ma$$

$$F_1 (\cos(\beta - \varphi) + k_t \sin(\beta - \varphi)) = ma + mg (\sin \varphi + k_t \cos \varphi)$$

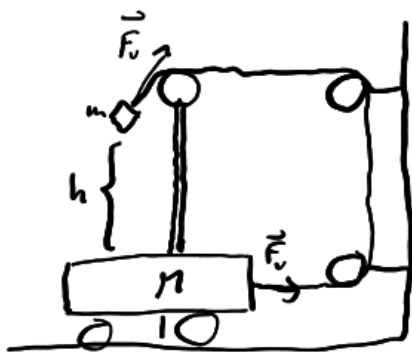
$$F_1 = \frac{ma + mg (\sin \varphi + k_t \cos \varphi)}{\cos(\beta - \varphi) + k_t \sin(\beta - \varphi)}$$

$$P = 50 \cdot F_1 \cdot a \cdot t \cdot \cos(\beta - \varphi)$$

$$P(1s) = 443,50 \text{ W} = 22 \text{ kW}$$

$$P(2s) = 886,50 \text{ W} = 44 \text{ kW}$$

94/95 1. kolokvij, (4)



$$M = 0,3 \text{ kg}$$

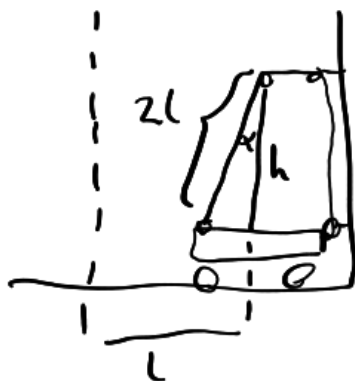
$$m = 0,15 \text{ kg}$$

$$k_t = 0$$

$$h = 0,2 \text{ m}$$

$$t = ?$$

$$2l = \frac{h}{\cos \alpha}$$



voziček



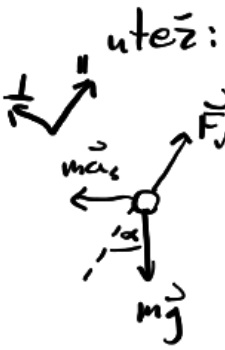
$$F_v + (F_v - F_v \sin \alpha) = 11a$$

$$F_v (2 - \sin \alpha) = 11a$$

$$\text{ker } s_u = 2s_v$$

$$a_u = 2a$$

$$\parallel: F_v - m\sqrt{a_s^2 + g^2} = -2ma$$



$$\vec{a}_s = -\vec{a}$$

$$(\vec{a}_s + \vec{g}) \parallel \vec{F}_v$$

$$\frac{a_s}{g} = \tan \alpha$$

$$F_v \left( 2 - \frac{a_s}{\sqrt{a_s^2 + g^2}} \right) = 11a$$

$$F_v = \frac{11a}{2 - \frac{a}{\sqrt{a^2 + g^2}}}$$

$$2ma = m\sqrt{a^2 + g^2} - \frac{11a\sqrt{a^2 + g^2}}{2\sqrt{a^2 + g^2} - a}$$

$$-11a\sqrt{a^2 + g^2} + m(a^2 + g^2) \cdot 2 - ma\sqrt{a^2 + g^2} = 4ma\sqrt{a^2 + g^2} - 2ma^2$$

$$\sqrt{a^2 + g^2} (5ma + 11a) = 4ma^2 + 2mg^2$$

$$(a^2 + g^2)(25m^2a^2 + 10mMa^2 + M^2a^2) = 4m^2(4a^4 + g^4)$$

$$a^4(9m^2 + 10mM + M^2) + a^2g^2(Ma + 5ma)^2 - 4g^4m^2 = 0$$

$$a = 0,33g$$

$$\alpha = 18,3^\circ$$

$$\frac{h}{\cos \alpha} = \frac{a_0 t^2}{2}$$

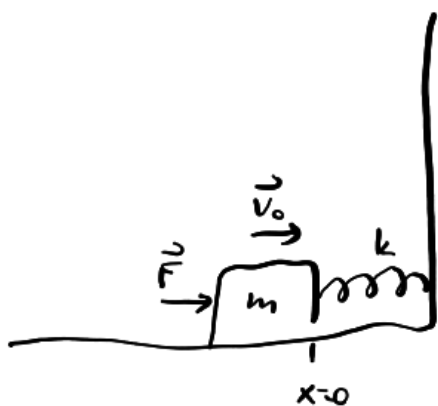
$$t = \sqrt{\frac{2h}{2a \cos \alpha}} = 0,25s$$

(3/27)

 $m, v_0, k, b$ 

$$v = v_0 - bx$$

$$x(P_{\max}) = ?$$



$$P = F \cdot v$$

$$F - kx = ma$$

$$a = \frac{dv}{dt} \frac{dx}{dx} = v \cdot \frac{dv}{dx}$$

$$F - kx = (v_0 - bx)(-b)m$$

$$F = (kx - (v_0 b + b^2 x)m) = x(k + b^2 m) - v_0 b m$$

$$P = (x(k + b^2 m) - v_0 b m)(v_0 - bx) =$$

$$= x v_0 (k + b^2 m) - v_0^2 b m - x^2 b (k + b^2 m) + v_0 b^2 m x =$$

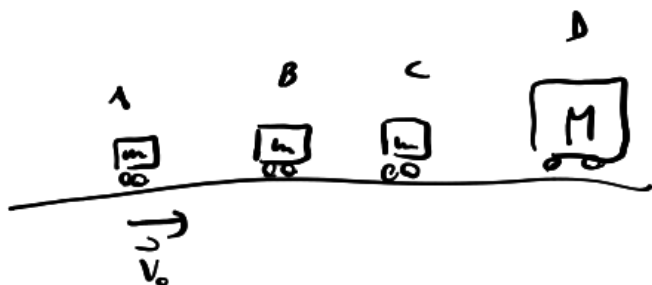
$$= x^2 (-bk - b^3 m) + x(v_0(k + b^2 m) - v_0 b^2 m) - v_0^2 b m$$

$$\frac{dP}{dx} = 2x(-bk - b^3 m) + v_0(k + b^2 m) - v_0 b^2 m = 0$$

$$x = \frac{v_0 k + 2v_0 b^2 m}{2(-bk - b^3 m)}$$

če še enkrat dvajamo, vidimo, da je to maksimum

Gibalna količina



$$m = 2 \text{ kg}$$

$$M = 3 \text{ kg}$$

a) 1. trk:

$$m v_0 = (m + m) v_1$$

$$v_0 = 5 \frac{\text{m}}{\text{s}}$$

a) nepr. trk

b) pr. trk

$$v_k = ?$$

$$v_1 = \frac{v_0}{2}$$

2. trk:

$$2mv_1 = 3mv_2$$

$$v_2 = \frac{2}{3}v_1 = \frac{v_0}{3}$$

3. trk:

$$3mv_2 = (3m+M)v_k$$

$$v_k = \frac{mv_0}{(3m+M)} = \frac{10}{(6+3)} = 1,1 \frac{\text{m}}{\text{s}}$$

b) 1. trk:

$$mv_0 = mv_A + mv_B$$

$$mv_0^2 = mv_A^2 + mv_B^2$$

$$\div \begin{cases} m(v_0 + v_A)(v_0 - v_A) = mv_B^2 \\ m(v_0 - v_A) = mv_B \end{cases}$$

$$v_0 + v_A = v_B$$

$$v_0 = v_A + v_B$$

$$v_A = 0$$

$$\boxed{v_B = v_0}$$

2. trk:

(podobno)

$$v_{Bk} = 0$$

$$v_c = v_0$$

3. trk:

$$mv_0 = mv_{ck} + Mv_d$$

$$mv_0^2 = mv_{ck}^2 + Mv_d^2$$

$$v_0 + v_{ck} = v_d$$

$$m(v_0 - v_{ck}) = M(v_0 + v_{ck})$$

$$v_{ck}(M+m) = v_0(m-M)$$

$$v_{ck} = v_0 \frac{m-M}{m+M} = 5 \frac{-1}{5} = -1 \frac{\text{m}}{\text{s}}$$

$$v_d = 5 - 1 = 4 \frac{\text{m}}{\text{s}}$$

29/35

$$M = 150 \text{ kg}$$

$$m = 70 \text{ kg}$$

$$v_m = 5 \text{ m/s}$$

$$\alpha = 30^\circ$$

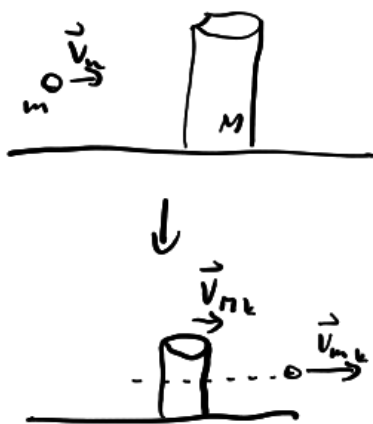
$$v_k, F_{\text{ot}_{\text{tir}}} = ?$$

$$F_{\text{ot}_{\text{tir}}} = m v_m \sin \alpha = 1,4 \text{ Ns}$$

$$m v_m \cos \alpha = (m + M) v_k$$

$$v_k = \frac{m v_m \cos \alpha}{m + M} = 1,4 \frac{\text{m}}{\text{s}}$$

29/33



$$2r = 10 \text{ cm}$$

$$M = 1 \text{ kg}$$

$$m = 2 \text{ g}$$

$$v_m = 300 \frac{\text{m}}{\text{s}}$$

$$F = 500 \text{ N} \quad \text{— sila valja na metek}$$

$$v_{mk}, v_{mk} = ?$$

$$m v_m = m v_{mk} + M v_{mk}$$

$$A = F 2r = 50 \text{ J}$$

$$W_m = W_{mk} + A + W_{mk}$$

$$\sqrt{\frac{m v_m^2 - m v_{mk}^2 - 2A}{M}} = v_{mk}$$

$$M (m v_m^2 - m v_{mk}^2 - 2A) = m^2 (v_m - v_{mk})^2$$

$$-M m v_{mk}^2 + M m v_m^2 - 2A M = m^2 v_m^2 - 2m^2 v_m v_{mk} + m^2 v_{mk}^2$$

$$v_{mk}^2 (m^2 + M m) + v_{mk} (-2m^2 v_m) + (m^2 v_m^2 + 2A m - M m v_m^2) = 0$$

$$V_{mk} \underbrace{\left(1 + \frac{1}{M}\right)}_{\downarrow 1} + V_{mk} \underbrace{\left(-2V_{mk} \frac{m}{M}\right)}_{\downarrow 0} + \left(V_{mk}^2 \underbrace{\left(\frac{m}{M} - 1\right)}_{\downarrow -1} + \frac{2A}{m}\right) = 0$$

$$\frac{m}{M} \rightarrow 0$$

$$V_{mk} = \pm \sqrt{V_{mk}^2 - \frac{2A}{m}} \approx \underline{\underline{200 \frac{m}{s}}}$$

$$V_{mk} = \frac{2(V_{mk} - V_{mk})}{M} \approx \underline{\underline{9,2 \frac{m}{s}}}$$

